

Flexible Specific Determination of Glucose in Solution, Blood Serum, and Sweat Using a Terahertz Hydrogel-Functionalized Metamaterial

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Terahertz metamaterials have shown promises in biomolecules detection, however, aqueous solvents such as blood serum strongly absorb the terahertz waves, interfering with the detection. To avoid such interference, an ultrathin, flexible, biogel-based metamaterial that can specifically detect the transmission of terahertz waves through biomolecules in aqueous solution is proposed and demonstrated. The glucose in water can be detected with a sensitivity of $0.0446 \text{ dL mg}^{-1}$ and a detection limit of 1.64 mg dL^{-1} , as well as glucose in human serum. The metamaterial that detects glucose in human sweat is part of a wearable device. The biosensor functions well for more than 100 detection cycles. The results suggest a strategy for real-time detection of glucose and potentially other biomolecules.

1. Introduction

Terahertz spectroscopy has shown promise, at least in vitro, for label-free detection of glucose, proteins, lactic acid, and amino acids.^[1–3] The extremely low energy of terahertz waves does not ionize biomolecules or otherwise perturb their structure.^[4–7] On the other hand, terahertz spectroscopy of biomolecules is limited by the fact that polar solvent molecules and their hydrogen bonding with one another and with biomolecules strongly absorb the terahertz waves, masking the signal from the target

biomolecules.^[8] This interference can be avoided in the solid state by using terahertz metamaterials,^[9–12] which are structures of subwavelength periodicity that show resonance and coupling peaks in the terahertz band. These structures excite local electric fields that enhance the interaction between terahertz waves and target biomolecules,^[13–15] allowing the detection of proteins and nucleic acids.^[16–18] Such detection requires drying and recrystallizing biological samples,^[19] which perturbs their structure and destroys their bioactivity.

We wondered whether it might be possible to exploit terahertz metamaterials for biomolecule detection in

native tissues by building them from hydrogels. Hydrogels are highly flexible, 3D networks of cross-linked polymers in water that can respond to external stimuli and swell without dissolving. Target biomolecules can react with components of the hydrogel to alter its water content and thereby thickness and dielectric constant,^[20,21] and terahertz waves can detect this variation.^[22] Hydrogels help preserve the normal structure and activity of biomolecules by maintaining them in an aqueous environment.^[23–26] So far, hydrogels have been designed to detect biomolecules in water or Tris buffer,^[27–29] nucleic acids in DEPC-treated water,^[30] viruses in Tris-borate-EDTA buffer,^[31] and glucose in phosphate-buffered saline (PBS) or buffer containing 2-(cyclo-hexylamino) ethane sulfonic acid (CHES), and NaOH.^[22,32,33] For example, a hydrogel based on 3-acrylamidephenylboric acid (AAPBA) can detect glucose in PBS or CHES/NaOH buffer,^[22,34] because glucose binds reversibly to the AAPBA, causing it to ionize and the hydrogel to swell.^[21,35]

The existing reports on glucose detection in the terahertz frequency range primarily utilize the terahertz attenuated total reflection (THz-ATR) method.^[36] Although this method exhibits high sensitivity for measuring liquids, it has notable limitations. The THz-ATR testing system is prone to random errors in amplitude and phase, as well as deviations in the incident angle of terahertz waves on the prism, which can significantly impact the extraction of sample information.^[37] Even a small prism misalignment in the order of a thousandth of a radian can produce a significant deviation in the extracted sample signal, the impacts of which far exceed the amplitude

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and phase errors in transmission-based terahertz time-domain spectroscopy (THz-TDS) systems.^[38] The generation of the total reflection phenomenon relies on the difference in refractive indices. For biological samples whose refractive index is close to the dielectric substrate, the measurement efficiency is reduced, and the ensuing multiple reflections may cause significant measurement errors.^[36] Furthermore, THz-ATR testing requires sample thickness greater than the skin depth, and excessively thick hydrogel samples lead to a long swelling equilibrium time. For instance, a 1 mm thick hydrogel film may require up to 2 days to achieve swelling equilibrium, severely affecting the practical convenience of detection applications.^[22] Compared to the THz-ATR system, the transmitted THz-TDS detection system can avoid the influence of prism misalignment, multiple reflections, as well as the instability of amplitude and phase, and allows for sample thickness reduction to tens to hundreds of micrometers. Traditional invasive glucose meters pose an increased risk of infection due to long-term multiple blood sampling when detecting glucose concentration. They only provide a single glucose concentration value after each measurement, lacking the ability to monitor blood glucose in real-time and continuously.^[39,40] In contrast, the non-invasive in vitro detection method of the terahertz detection system can effectively avoid the infection risk caused by blood sampling.^[40,41] This system is capable of penetrating the human body surface without causing ionizing radiation damage, offering the potential to achieve continuous detection of glucose concentration in vitro and improve the management of diabetes.^[42–45] However, for biological solutions, the large amount of moisture in the solution leads to the strong absorption of terahertz waves, making direct detection of biological solutions via transmission THz-TDS systems ineffective.^[46,47] To address this, there is an urgent need to develop a terahertz sensor that is extremely sensitive to minute changes in hydration state, aiming to balance the absorption of terahertz waves by water and shorten the swelling equilibrium response time, thereby improving detection efficiency of glucose biomolecules. In addition, such continuous, non-invasive detection could ideally be achieved if glucose-sensing hydrogels could be fabricated as flexible, wearable devices, in contrast to today's rigid hydrogel sensors.^[20,48]

Here we present an ultra-thin, flexible, AAPBA hydrogel that allows, for the first time, the detection of a biomolecule in an aqueous solution based on terahertz emission. A 200 μm -thick hydrogel film was overlaid onto the upper surface of a terahertz metamaterial consisting of an array of polarization-insensitive double split-ring resonators. The prepared terahertz metamaterials based on ultrathin hydrogel films significantly shorten the swelling equilibrium response time, thereby improving the detection efficiency. The metamaterial detected glucose in aqueous solution with a sensitivity of $0.0446 \text{ dL mg}^{-1}$ and a detection limit as low as 1.64 mg dL^{-1} , and the sensor could be recycled many times by precipitating detected glucose off the surface using CHES/NaOH buffer. The sensor showed excellent performance even after bending many times, and it detected glucose in sweat when attached to the skin. This novel sensor design provides a new direction for the development of flexible biosensors based on terahertz metamaterials.

2. Results and Discussion

2.1. Biosensor Design and Working Principle

The device consists of a flexible array of polarization-insensitive double split-ring resonators as well as an AAPBA hydrogel film (Figure 1a). The hydrogel metamaterial is transferred to a flexible, highly transmissive substrate of polyethylene terephthalate (Figure S1, Supporting Information). Each ring resonator in the unit cell has a gap of size $8 \mu\text{m}$ on all four walls, forming a structure with C_4 symmetry (Figure 1b). The Simulation predicted that the transmission spectra from the double-resonator design would be insensitive to the polarization of the incident radiation (Figure S2b, Supporting Information), reflecting its C_4 symmetry. This insensitivity means that detection should be robust to deviations between the position of the metamaterial and the polarization direction of the incident electromagnetic wave. When the incident radiation was linearly polarized, metamaterials with only one split ring resonator exhibited transmission dips at 0.7 and 0.9 THz (Figure S2a, Supporting Information). Bright-mode resonances due to free electrons in the metal surface occurred at the open gaps, where the electric field energy distribution was strongest (Figure S2c,d, Supporting Information). The bright-mode resonances generated by each single split-ring resonator constructively interfered with each other, leading to a transmission peak at 0.81 THz in the double resonator design (Figure 1c). The array of double split-ring resonators was prepared from aluminum (200 nm thick), which was coupled to an AAPBA hydrogel (200 μm thick) using UV curing, and etched by photolithography onto a flexible polyimide substrate (Figure S3, Supporting Information).

In an aqueous solution, the uncharged AAPBA reversibly forms borate esters with monosaccharides such as glucose,^[21,22,33,49,50] creating bulkier, negatively charged hydrogel components that cause the hydrogel to swell (Figure S4, Supporting Information). Before exposure to glucose, we first equilibrated the AAPBA-based hydrogels in CHES/NaOH buffer to ensure there was no more net absorption of water. We predicted that reequilibrating the hydrogel in CHES/NaOH after exposure to glucose would allow its reuse, since the buffer strips off bound glucose and returns the hydrogel to its original state.^[22,49]

2.2. Simulated and Experimental Validation of the Terahertz Metamaterial and Biosensor Design

We formulated the AAPBA hydrogels with different mass fractions of AAPBA, cross-linkers, and different thicknesses, then compared them in terms of various characteristics important for glucose sensing (Figure S5, Supporting Information). We used the optimized hydrogel formulation of 9% AAPBA, 1.4% cross-linker, and thickness of 200 μm in all subsequent experiments. The simulated and experimental transmission spectra of the optimized hydrogel metamaterial showed a distinct resonance-coupled transmission peak $\approx 0.88 \text{ THz}$ (Figure 2a), and the transmittance was much higher than that reported for a thicker hydrogel (1 mm) containing the same concentrations of AAPBA and cross-linker as our hydrogel.^[22] The hydrogel film took only $\approx 25 \text{ min}$ to achieve swelling equilibrium at room temperature

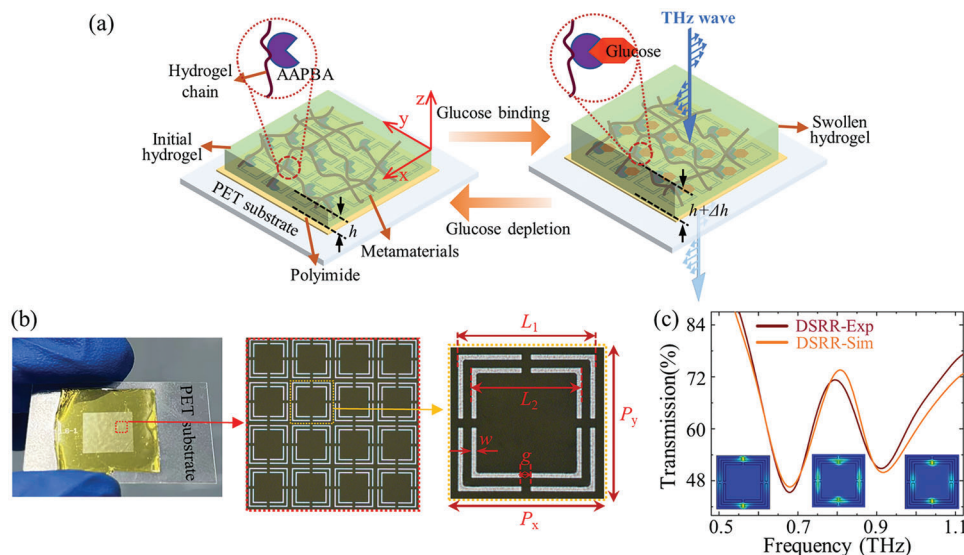


Figure 1. Working principle and characteristics of a terahertz metamaterial biosensor based on an AAPBA hydrogel. a) Schematic diagram of the biosensor, which swells upon glucose binding. b) Photograph of the biosensor (measuring $7.8 \times 7.8 \text{ mm}^2$) fabricated on the yellow polyimide substrate ($20 \mu\text{m}$ thick), and optical micrograph of the biosensor. The schematic at right shows a unit cell with the following characteristics: $P_x = P_y = 173 \mu\text{m}$, $L_1 = 159 \mu\text{m}$, $L_2 = 129 \mu\text{m}$, $g = w = 8 \mu\text{m}$. AAPBA, 3-acrylamidephenylboric acid; PET, polyethylene terephthalate. c) Experimental (Exp) and simulated (Sim) transmission spectra of terahertz metamaterial with double split-ring resonators (DSRR) on the outside and inside. Transmission is the function of terahertz frequency. The insets are the electric field enhancement at the single split-ring resonator gap on the outside (outer-SSR) at resonance frequency 0.7 THz , at the DSRR gap at coupling frequency 0.81 THz , and at the single split-ring resonator gap on the inside (inner-SSR) at resonance frequency 0.9 THz .

(Figure 2b), reflecting its thinness.^[51] Indeed, this equilibration time is shorter than the 60 min reported for a hydrogel $350 \mu\text{m}$ thick containing 20% AAPBA and 1.5% cross-linker.^[21]

2.3. Detection of Glucose in Aqueous Solution

The AAPBA hydrogel film on its own did not show transmission or absorption peaks after equilibration in glucose solution, whereas the hydrogel-based metamaterial showed a clear resonance-coupled transmission peak $\approx 0.88 \text{ THz}$ (Figure 3a; Figure S6, Supporting Information). The transmission from the hydrogel metamaterial decreased with increasing glucose

concentration (Figure 3b), reflecting the increasing swelling due to the reaction between glucose and AAPBA. For glucose concentrations up to 100 mg dL^{-1} , the transmission peak of the hydrogel metamaterial varied linearly, and detection sensitivity was $S_1 = 0.0446 \text{ dL mg}^{-1}$, corresponding to 0.803 mm^{-1} (Figure 3c,d). This sensitivity is around twice that reported for hydrogel ($350 \mu\text{m}$ thick) containing 20% AAPBA and 1.5% cross-linker.^[21] The detection limit of our biosensor was 1.64 mg dL^{-1} (corresponding to 0.09 mm), compared to the limits of $10\text{--}17.72 \text{ mg dL}^{-1}$ reported in the literature.^[20,22] At glucose concentrations from 100 to 300 mg dL^{-1} , the sensitivity of our biosensor fell to $0.0144 \text{ dL mg}^{-1}$ (corresponding to 0.26 mm^{-1}). This decrease may reflect that AAPBA sites available to react with

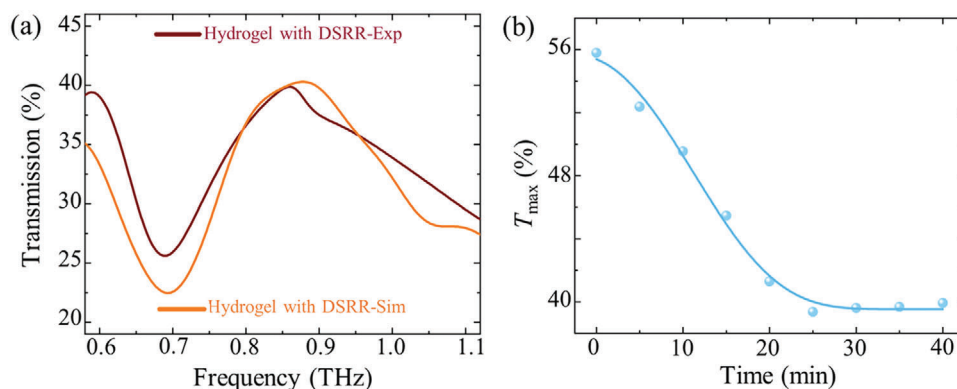


Figure 2. Characterization of the terahertz metamaterial biosensor based on an AAPBA hydrogel. a) Experimental (Exp) and simulated (Sim) transmission spectra of the hydrogel prepared on the DSRR metamaterial (as described in Figure S5d, Supporting Information). b) Maximal transmission of the hydrogel prepared on the DSRR metamaterial (as described in Figure S5d, Supporting Information) as a function of time in CHES/NaOH buffer. Equilibration was achieved after $\approx 25 \text{ min}$. AAPBA, 3-acrylamidephenylboric acid; CHES, 2-(cyclo-hexylamino) ethane sulfonic acid; EGDM, ethylene glycol dimethacrylate.

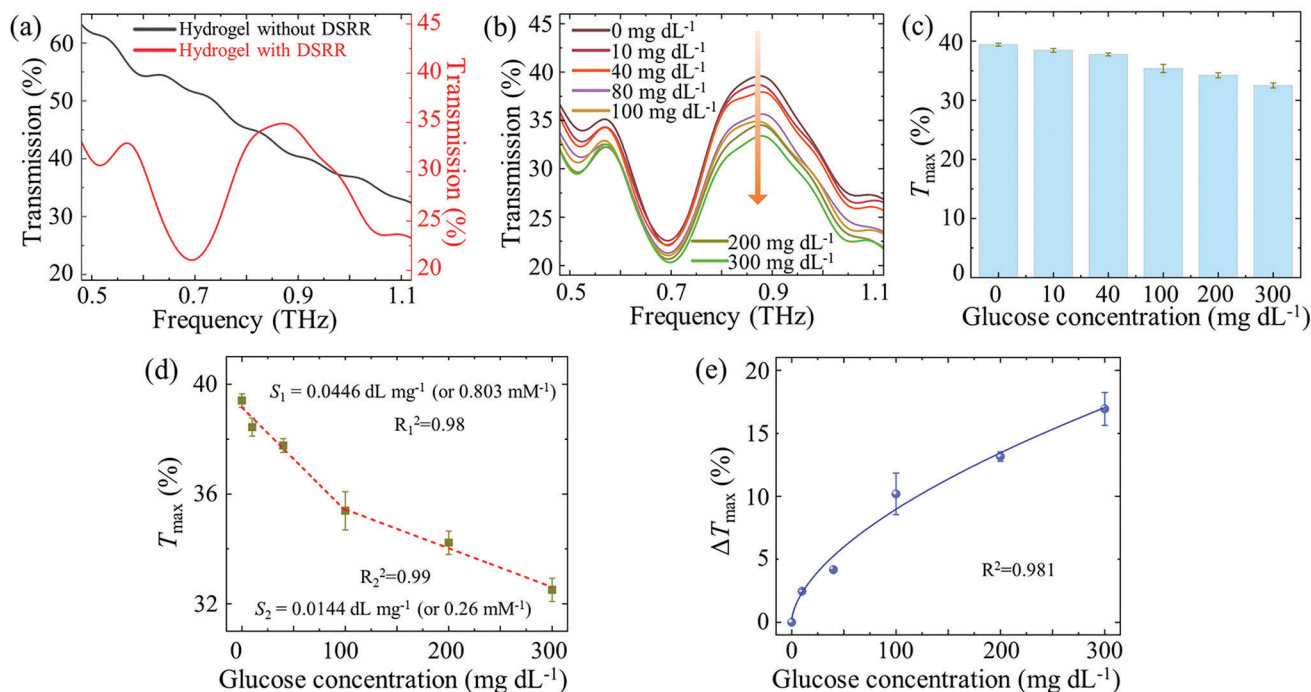


Figure 3. Detection of glucose in aqueous solution. Terahertz transmission spectra were collected for a) hydrogel films on their own or coupled with the double split-ring resonators (DSRR), after equilibration in an aqueous solution containing 100 mg dL⁻¹ glucose; or b) hydrogel-DSRR, after equilibration in aqueous solutions containing the indicated glucose concentrations. c) Variation of maximal transmission as a function of glucose concentration. d) Biphasic dependence of maximal transmission on glucose concentration, where the slope of each phase indicates the sensitivity S (see Experimental Section). e) Normalized variation in terahertz transmission as a function of glucose concentration. Error bars indicate the standard deviation ($n = 3$).

glucose and the hydrogel's intrinsic elasticity becomes limiting.^[21] Nevertheless, the normalized signal variation ($\Delta T_{\max}$) remained high even at higher glucose concentrations (Figure 3e). Furthermore, the glucose detection performance of the proposed hydrogel metamaterial sensor has been compared with previously proposed optical glucose sensors (as shown in Table S1, Supporting information). The hydrogel terahertz metamaterial sensor we designed exhibited various degrees of superiority in terms of glucose specificity detection, response time, detection sensitivity, and detection limit.

The biosensor gave consistent responses even after up to 100 cycles of swelling and deswelling with CHES/NaOH buffer (Figure S7, Supporting Information). The hydrogel has highly reversible swelling–shrinking behavior and long-term stability of signal responsiveness. Compared to the typical glucose oxidase-based detection method, it has no drawbacks in large signal drifts and immunological rejection with increasing repetition.^[33] Our hydrogel metamaterial sensor reveals a more applicable sensing strategy with good biocompatibility and reusability. Subsequently, we investigated the stability of the hydrogel metamaterial biosensor. Figure S7c (Supporting Information) shows that after long-term storage at room temperature (80 days), the transmission coefficient of hydrogel metamaterial biosensor incubated in the CHES/NaOH buffer did not change significantly. This result emphasizes the high stability profiting from the synthesized hydrogels and further confirms that the cause of the transmittance variation is indeed the glucose-induced hydration state shift. It also confirms the saturated swelling degree of the hydrogel metamaterial biosensor.

It also gave consistent responses after up to 80 bending treatments (Figure 4a), with no extensive deformation or cracking observed. Other monosaccharides such as fructose and galactose, which can also react with AABPA,^[20] interfered minimally with the detection of glucose (Figure 4b–d). The degree of interference (see Experimental Section) was only 2.52% for fructose, 1.26% for galactose. The presence of substances such as uric acid, lactic acid, and hyaluronic acid might theoretically affect the measurement results. However, these influences are usually small in practical applications. Studies have shown that when the interference degree is less than 5%, the influence of these interferences on glucose concentration detection can be ignored.^[52] Our experimental results show that the interference degree was 2.58% for hyaluronic acid (HA), 1.54% for uric acid (UA), and 4.9% for lactic acid (LA), all of which were lower than 5%. This implies negligible interference when detecting glucose in humans, whose blood contains 70–124 mg dL⁻¹ glucose in a healthy state or 129–180 mg dL⁻¹ in the presence of diabetes, compared to <1.8 mg dL⁻¹ fructose, galactose or HA, and ≤12 mg dL⁻¹ UA or LA.^[21,53,54] Adding to the biosensor's specificity is that large biomolecules such as proteins are unlikely to adsorb in the hydrogel because of the abundant hydroxyl groups in the gel network and the sieving properties of the network itself.^[22]

Since our ultimate goal is non-invasive glucose monitoring and we therefore planned to measure its levels in sweat (see below), we also examined to what extent inorganic ions in human sweat might interfere with glucose determination. We equilibrated the hydrogel in an aqueous solution containing

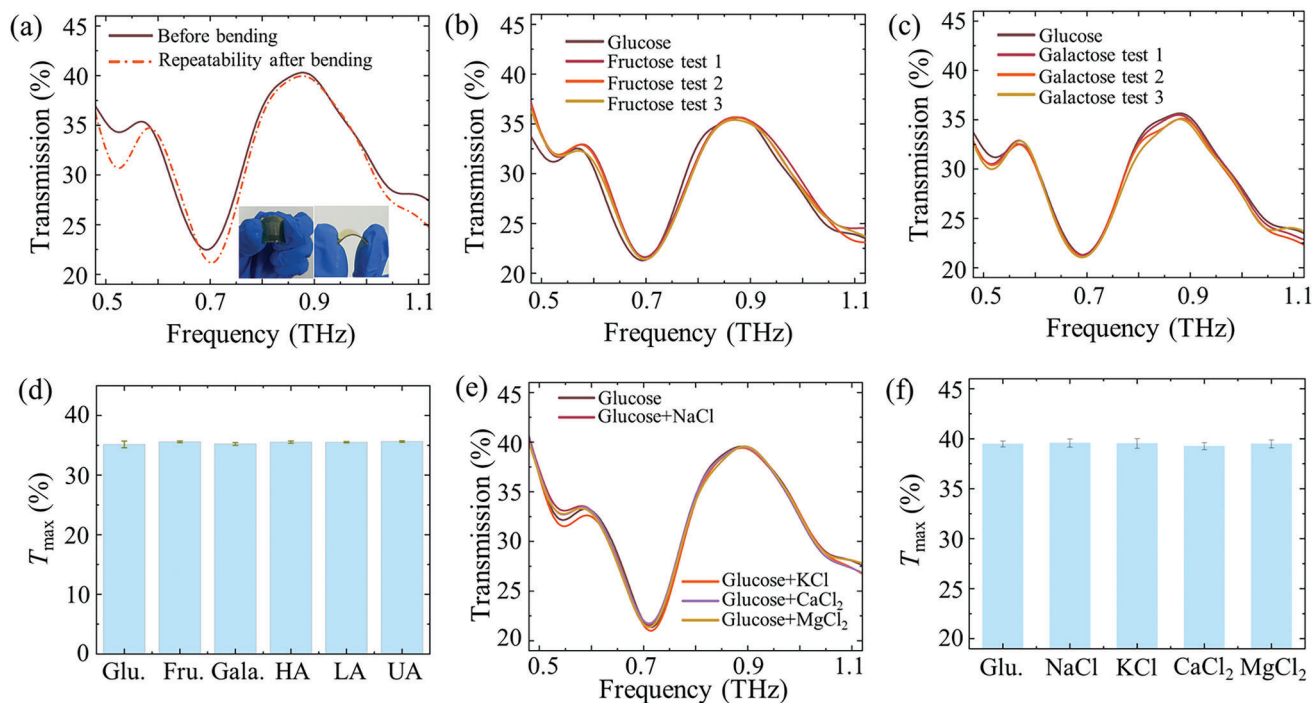


Figure 4. Robustness of glucose detection in aqueous solution to biosensor bending and interference from other monosaccharides. a) The terahertz transmission after each bending was compared with that before any bending, and the inset is a photograph of hydrogel-based metamaterial being bent by 45°. This bending motion was performed up to 80 times. In all cases, the hydrogel was equilibrated in CHES/NaOH buffer. b,c) Terahertz transmission spectra of hydrogels equilibrated in aqueous solutions of 80 mg dL⁻¹ glucose alone or mixed with 2 mg dL⁻¹ fructose or with 2 mg dL⁻¹ galactose. Three independent tests with the fructose- or galactose-containing solutions are shown. d) Maximal transmission under the six conditions of fructose or galactose, HA, LA, and UA-containing glucose solutions. e,f) Comparison of terahertz transmission spectra from the hydrogel equilibrated in aqueous solutions containing 3.6 mg dL⁻¹ glucose alone or together with one of the following inorganic salts: NaCl, 200 mg dL⁻¹; KCl, 38 mg dL⁻¹; CaCl₂, 4 mg dL⁻¹; or MgCl₂·6H₂O, 2 mg dL⁻¹.

3.6 mg dL⁻¹ glucose by itself or mixed with one of the following inorganic salts at concentrations higher than their physiological levels in sweat:^[55] Na⁺ at 200 mg dL⁻¹ (vs 90 mg dL⁻¹ in normal sweat), K⁺ at 38 mg dL⁻¹ (vs 20 mg dL⁻¹), Ca⁺ at 4 mg dL⁻¹ (vs 1.5 mg dL⁻¹), or Mg⁺ at 2 mg dL⁻¹ (vs 0.13 mg dL⁻¹). None of these salts substantially interfered with glucose determination (Figure 4e,f).

2.4. Detection of Glucose in Humans

Next, we evaluated the accuracy of our biosensor in the detection of glucose in human serum. The concentrations determined from the serum of three healthy individuals were extremely close to those determined using a commercial glucometer (Figure 5a). To permit entirely non-invasive glucose monitoring, we examined the ability of our biosensor to assay glucose in sweat. Glucose concentration in human sweat ranges from 0.18 to 19.98 mg dL⁻¹, much lower than the concentration in blood.^[56,57] Individuals with diabetes secrete larger concentrations of glucose into their sweat,^[58] and the concentration in sweat increases linearly with its concentration in blood.^[59,60] The pH of human sweat, 6.3, is much more acidic than the pK_a of AAPBA (8.2),^[61] which we feared might inhibit the deprotonation of hydroxyl groups in the adduct that forms between glucose and AAPBA, thereby inhibiting hydrogel swelling. Therefore, we first

equilibrated the hydrogel in CHES/NaOH buffer at pH 9.0, then applied it to the back of the subject's hand. The glucose measurements began to stabilize just a few minutes after sweating was induced (Figure 5b,c). In addition, the glucose concentrations in sweat mirrored the initial rise and subsequent decrease of glucose concentration in blood (Figure 5d). The deviations between sweat glucose measurements from our biosensor and blood glucose measurements from a commercial glucometer may reflect, in part, the deposition of dead epidermis tissue and other skin contaminants onto the biosensor.^[58]

During real-time monitoring of human sweat glucose concentration (Figure 5c), the time interval between the two measurements is very short, and the hydrogel metamaterial biosensor needs to rapidly reset to the skin after each measurement to confirm the time to reach a stable detection of sweat glucose concentration. Therefore, there is no need to undergo the recovery cycles in between the measurements. In the case of Figure 5d, where measurements are carried out before, during, and after meals with a longer interval, the porous structure of the hydrogel could potentially absorb impurities or other chemicals from the air, which may interfere with subsequent test results. Therefore, we placed hydrogel metamaterial biosensor in CHES/NaOH buffer between each measurement to undergo a recovery cycle. To purify the impurities in the hydrogel and restore its chemical environment, thereby ensuring the accuracy and repeatability of the measurements.

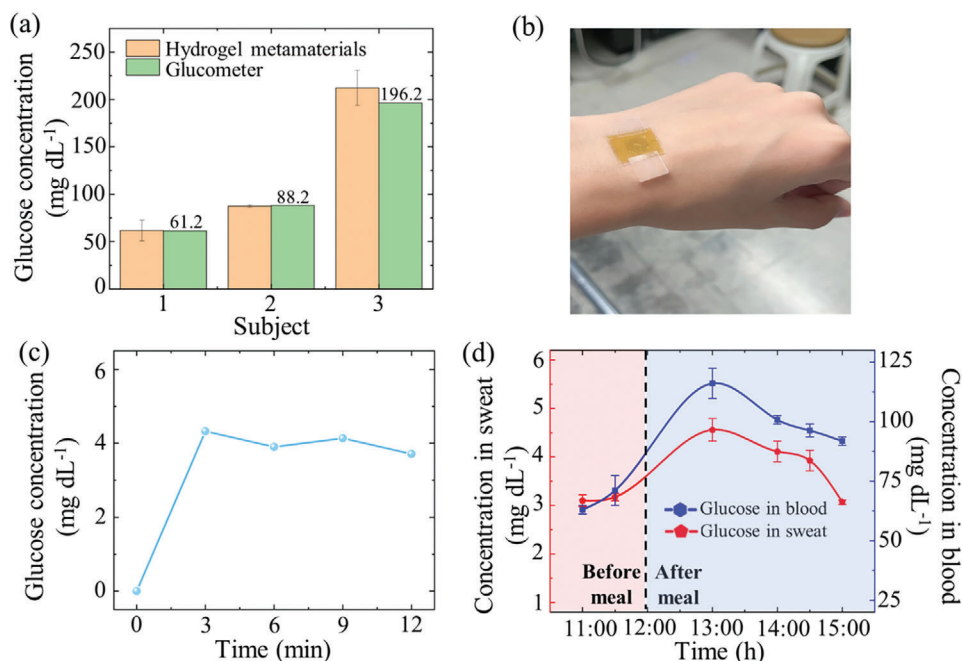


Figure 5. Glucose detection in human serum and sweat. a) Comparison of glucose concentrations in the serum of three healthy individuals as determined using our biosensor (yellow) or a commercial blood glucose meter (green). b) Photograph of the biosensor attached to a subject's hand. c) Subjects were asked to exercise, wear a glove, or wear heating pads on the skin of arms in order to induce sweating.^[58,62] Representative real-time measurement of glucose during sweating of one subject. d) Comparison of glucose concentrations before and after a meal, as measured in sweat using our biosensor (red) or in blood using a commercial glucometer (blue).

3. Conclusion

We have demonstrated an ultrathin, flexible terahertz metamaterial based on AAPBA hydrogel that can transmissively detect glucose in aqueous solution, human serum, and human sweat. Compared to the ATR system, the transmitted TDS detection effectively avoids the influence of prism displacement, multiple reflections, as well as the instability of amplitude and phase during biological sample testing. It also allows the hydrogel thickness to be reduced to 200 μm , enabling a rapid response time. With detection sensitivity as high as $0.0446 \text{ dL mg}^{-1}$ in the range of $0\text{--}100 \text{ mg dL}^{-1}$, and a detection limit as low as 1.64 mg dL^{-1} . The influence of fructose, galactose, HA, LA, UA, or inorganic ions interferences on glucose concentration detection in humans can be ignored. The detector can be used multiple times without loss of performance. Our system allows non-invasive, real-time detection of glucose as a wearable device, opening up new possibilities for the long-term management of diabetes. Our strategy may be extendable to other biomolecules, creating new possibilities for terahertz metamaterial biosensing.

4. Experimental Section

Reagents: The following reagents were purchased from Sigma-Aldrich (Milwaukee, WI, USA): 2-hydroxyethyl methacrylate (HEMA), 2,2-dimethoxy-2-phenylacetophenone (DMPA), and ethylene glycol dimethacrylate (EGDM). Human serum was purchased from Solarbio (Beijing, China), while 3-acrylamidophenylboric acid (AAPBA) was purchased from Aladdin (Shanghai, China). The buffer to strip glucose off the AAPBA hydrogel was prepared by mixing 2-(cyclo-hexamino) ethane

sulfonic acid (CHES, Solarbio, 1.555 g dL^{-1}) and NaCl (0.707 g dL^{-1}), then adjusting the pH to 7.4 or 9.0 using NaOH (1 M). The CHES/NaOH buffer at pH 9.0 was used to equilibrate AAPBA hydrogels, while the buffer at pH 7.4 was used to prepare aqueous solutions of glucose alone or together with fructose, galactose, hyaluronic acid, uric acid, and lactic acid, or inorganic salts.

Synthesis of AAPBA Hydrogel Films on Their Own: To prepare hydrogels containing a 9% mass fraction of AAPBA and 1.4% EGDM as cross-linker, AAPBA (90 mg), DMPA (36 mg), and EGDM (14 mg) were dissolved in the solvent HEMA (860 mg) and mixed together in a microcentrifuge tube wrapped in aluminum foil. After mixing, the tube was incubated at 37°C in the dark for 30 min, until a clear "hydrogel precursor" solution was obtained. The amounts of AAPBA and EGDM were adjusted to prepare hydrogel precursor solutions with the formulations indicated in the figures.

Hydrogel precursor solution (30 μL) was dropped onto a Teflon mold with a diameter of 1 cm and thickness of either 200 μm or 1 mm, then cured for 2.5 or 4 min, respectively, under a UV curing lamp (20 mW, $\lambda = 365 \text{ nm}$). The mold was soaked overnight in deionized water, then the hydrogel was carefully removed from the mold and soaked in CHES/NaOH buffer at pH 9.0 until it achieved a constant weight. Hydrogels were then incubated in aqueous solutions containing glucose alone or together with other biomolecules as described in the figures, and the degree of swelling was calculated as:

$$SD = \frac{W_c - W_0}{W_0} \quad (1)$$

where W_0 and W_c refer to the weights measured after equilibration in CHES/NaOH buffer or the corresponding aqueous solution, respectively.

Fabrication of Flexible Terahertz Metamaterials Containing AAPBA Hydrogel Films: A polyimide layer (20 μm thickness) was spin-coated onto a silicon sheet, and a layer of aluminum (200 nm thickness) was coated onto the polyimide surface using electron beam evaporation. A photoresist layer (500 nm thickness) was spun-coated onto the aluminum, and a

mask plate with periodic DSRR arrays was placed on top of the photore-sist, which was subjected to photolithography followed by wet etching. Hydrogel precursor solution (30 μL), prepared as described above, was added to the upper surface of the DSRR metamaterial, then a copper foil spacer (200 μm thickness) and a central opening was added, followed by silane-treated glass slide. The entire assembly was cured for 2.5 min under a UV curing lamp (20 mW), then the glass slide and copper foil spacers were removed, leaving a hydrogel (200 μm thickness) on the surface of the metamaterial. The metamaterial was carefully removed from the silicon sheet and attached to a flexible PET film (225 μm thickness) (see Figure S3, Supporting Information for details).

Simulations of Terahertz Transmission of AAPBA Hydrogels on Their Own or in Metamaterials: Simulations were conducted using Computer Simulation Technology Microwave Studio (version 2019, CST, Darmstadt, Germany). A polyimide layer (20 μm thickness) was assigned a dielectric constant of 3.1 and a loss tangent delta of 0.088. A hydrogel (200 μm thickness) was assigned a dielectric constant of 3.7 and a loss tangent delta of 0.2. A single, linearly polarized incident wave was specified perpendicular to the metamaterial surface.

Experimental Measurement of Terahertz Transmission of AAPBA Hydrogel Metamaterials: Fabricated metamaterials were characterized using transmission-type terahertz time-domain spectroscopy, as shown in Figure S8, (Supporting Information). The femtosecond laser oscillator (Coherent) used for the broadband terahertz excitation had a repetition rate of 80 MHz and spectral bandwidth of 70 nm. The pulsed laser beam with a central wavelength of 800 nm was focused onto a photoconductive switch. The beam waist diameter of the terahertz beam (3 mm) was much smaller than the periodic array on the sample, thus ensuring measurement validity.^[63,64] This system had a Fourier-transformed bandwidth of 0.1–2.5 THz. Before measurement, the metamaterials were equilibrated in CHES/NaOH buffer at pH 9.0, then overlaid with aqueous solutions (150 μL) of glucose alone or together with other biomolecules or inorganic salts. The excess solution sample on the metamaterials was removed by blotting with dust-free paper, then the terahertz transmission spectrum was recorded.

Measurement of Glucose in Human Sweat and Blood: All experiments involving human participations were approved by the Ethics Committees of Tianjin University (approval number: TJUE-2023-172). Informed written consent from all human participants was obtained prior to the experiments.

The skin on an arm or hand was first cleaned with soap, water, and medicinal alcohol. The hydrogel metamaterial biosensor was equilibrated in CHES/NaOH buffer at pH 9.0, then applied to the back of the arm or hand. Sweating was induced through exercising, wearing a glove, or wearing heating pads on the skin, then the biosensor was removed a few minutes later and attached to a sample stand with an aperture of 10 mm within the transmission-type terahertz time-domain spectroscopy system (Figure S8, Supporting Information). In these experiments, sweating was induced either before or after the subject ate a regular meal in order to obtain pre- and post-meal glucose measurements. During these experiments, glucose concentration in blood sampled from the fingertip was measured using a commercial glucometer (Yuyue, Jiangsu, China) before and after eating (Figure S9, Supporting Information).

Glucose Sensitivity of the Hydrogel Metamaterials: Sensitivity was defined as:

$$S = \frac{\Delta T}{\Delta C} = \frac{T_{0\max} - T_{c\max}}{C_c - C_0} \quad (2)$$

where $\Delta T = T_{0\max} - T_{c\max}$ is the change in the terahertz transmission peak. $T_{0\max}$ and $T_{c\max}$ refer to the transmission peaks at 0.88 THz measured after equilibration in CHES/NaOH buffer or glucose solution, respectively. $\Delta C = C_c - C_0$ is the change in glucose concentration, where C_0 refers to water without glucose and C_c to the aqueous solution of glucose. The normalized signal variation of the terahertz transmission coefficient was calculated as $\Delta T_{\max} = (T_{0\max} - T_{c\max})/T_{0\max}$.

Interference with Glucose Detection: The degree of interference from interfering substances was quantified using the formula:

$$DI = \frac{T_{80\max} - T_{I\max}}{T_{0\max} - T_{80\max}} \quad (3)$$

where $T_{80\max}$ and $T_{I\max}$ refer to the transmission peak of the hydrogel metamaterials at 0.88 THz, measured after equilibration in an aqueous solution containing 80 mg dL⁻¹ glucose alone or with the indicated interferent.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interests.

Data availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

glucose sensing, hydrogels, terahertz metamaterials, wearable biosensors

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